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Inventor(s)	Tisch; Erich et al.

Apparatus and method of removing selective sounds from a video

Abstract

An apparatus including a microphone, a speaker, and a processor. The speaker is configured to produce a sound that audibly communicates information to a user. The microphone is configured to receive the sound. The processor is configured to produce the sound through the speaker to audibly communicate information to the user; capture a video that includes audio; initiate a sound removal process to remove the sound from the audio; and after the sound is produced, stop the sound removal process.

Inventors: **Tisch; Erich (San Francisco, CA), Finkel; Paul (Redwood City, CA)**

Applicant: **GoPro, Inc.** (San Mateo, CA)

Family ID: **1000008779997**

Assignee: **GoPro, Inc. (San Mateo, CA)**

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Primary Examiner: Hunter; Mishawn N.

Attorney, Agent or Firm: Young Basile Hanlon & MacFarlane, P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation of U.S. patent application Ser. No. 17/979,017, filed Nov. 2, 2022, which claims priority to and the benefit of U.S. Provisional Application Patent Ser. No. 63/275,909, filed Nov. 4, 2021, the entire disclosure of which is hereby incorporated by reference.

TECHNICAL FIELD

(1) This disclosure relates to an image capture device that provides sounds indicating that a video is being taken and a method and apparatus for removing the sounds from the video.

BACKGROUND

(2) Generally, image capture devices are available that are capable of capturing both images and videos. These image capture devices may provide a sound indicating that an image or video is about to be taken such that the user or group of participants are alerted that an image or video is about to be captured.

SUMMARY

(3) Disclosed herein are implementations of an apparatus including a microphone, a speaker, and a processor. The speaker is configured to produce a sound that indicates an image capture device comprising the microphone is recording. The microphone is configured to receive the sound. The processor is configured to initiate a sound removal process to remove the sound; produce the sound through the speaker to indicate that the image capture device is recording a video that includes audio; and after the sound is produced, stop the sound removal process.

(4) The present teachings provide a method that includes initiating a sound and removing the sound. The method includes initiating the sound removal process. After initiation of the sound removal process, the method includes producing the sound that is indicative of an image capture device recording a video that includes audio. Upon completion of producing the sound, the method includes stopping the sound removal process.

(5) The present teachings provide an image capture device comprising: a speaker, a microphone, and an image sensor. The speaker is configured to produce a sound indicating that the image capture device is recording. The microphone configured to receive the sound. The image sensor configured to capture a video while the microphone is recording. A processor configured to: initiate a sound removal process to remove the sound from the recording. The processor configured to produce the sound to indicate that the image sensor and microphone are recording a video that includes audio. The processor configured to stop the sound removal process after a predetermined amount of time.

(6) The present teachings provide: an apparatus including a microphone, a speaker, and a processor. The speaker is configured to produce a sound that audibly communicates information to a user. The microphone is configured to receive the sound. The processor configured to: produce the sound through the speaker to audibly communicate information to the user; capture a video that includes audio; initiate a sound removal process to remove the sound from the audio; and after the sound is produced, stop the sound removal process.

(7) The present teachings provide: a method including initiating a sound removal process. After initiation of the sound removal process, producing a sound that audibly communicates information to a user. Upon completion of producing the sound, stopping the sound removal process.

(8) The present teachings provide: a image capture device with a speaker, a microphone, an image sensor, and a process. The speaker is configured to produce a sound that audibly communicates information to a user. The microphone is configured to receive the sound. The image sensor is configured to capture a video while the microphone is recording. The processor is configured to: initiate a sound removal process to remove the sound from the recording; produce the sound to audibly communicate information; and stop the sound removal process after a predetermined amount of time.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure is best understood from the following detailed description when read in conjunction with the accompanying drawings. It is emphasized that, according to common practice, the various features of the drawings are not to-scale. On the contrary, the dimensions of the various features are arbitrarily expanded or reduced for clarity.

(2) FIGS. **1**A-**1**B are isometric views of an example of an image capture device.

(3) FIGS. **2**A-**2**B are isometric views of another example of an image capture device.

(4) FIG. **3** is a block diagram of electronic components of an image capture device.

(5) FIG. **4**A is a block diagram of processing a recording to remove a sound.

(6) FIG. **4**B is a block diagram of processing a recording to remove a sound.

(7) FIG. **5**A illustrates a sound captured by a microphone.

(8) FIG. **5**B illustrates a filter to remove a sound captured by the microphone.

(9) FIG. **5**C illustrates a resulting signal after the sound is removed.

(10) FIG. **6** illustrates removing a sound when the microphones are not equidistant from a speaker.

DETAILED DESCRIPTION

(11) The present teachings provide an image capture device that captures videos and audio related to the video. The image capture device includes speakers and microphones. The speakers may produce an audible sound (e.g., a beep) so that a user of the device knows the device is recording when the device is in a location where the user is not able to monitor the recording status (e.g., on a helmet, a headband, or a backpack). The device taught herein may filter out the sound being produced so that when the video is watched later, the sound is not part of the audio recording. The device may include a processor that removes the sound from an audio recording, before the audio is stored, or both. The processor is configured to apply sound removal so that the sound is filtered out while the audio and video are being recorded or filtered out before an audio recording is made or completed. The processor is configured to perform a process of removing the sound without disrupting the audio being recorded.

(12) FIGS. **1**A-**1**B are isometric views of an example of an image capture device **100**. The image capture device **100** may include a body **102**, a lens **104** structured on a front surface of the body **102**, various indicators on the front surface of the body **102** (such as light-emitting diodes (LEDs), displays, and the like), various input mechanisms (such as buttons, switches, and/or touch-screens), and electronics (such as imaging electronics, power electronics, etc.) internal to the body **102** for capturing images via the lens **104** and/or performing other functions. The lens **104** is configured to receive light incident upon the lens **104** and to direct received light onto an image sensor internal to the body **102**. The image capture device **100** may be configured to capture images and video and to store captured images and video for subsequent display or playback.

(13) The image capture device **100** may include an LED or another form of indicator **106** to indicate a status of the image capture device **100** and a liquid-crystal display (LCD) or other form of a display **108** to show status such as battery life, camera mode, elapsed time, and the like. The image capture device **100** may also include a mode button **110** and a shutter button **112** that are configured to allow a user of the image capture device **100** to interact with the image capture device **100**. For example, the mode button **110** and the shutter button **112** may be used to turn the image capture device **100** on and off, scroll through modes and settings, and select modes and change settings. The image capture device **100** may include additional buttons or interfaces (not shown) to support and/or control additional functionality.

(14) The image capture device **100** may include a door **114** coupled to the body **102**, for example, using a hinge mechanism **116**. The door **114** may be secured to the body **102** using a latch mechanism **118** that releasably engages the body **102** at a position generally opposite the hinge

mechanism **116**. The door **114** may also include a seal **120** and a battery interface **122**. When the door **114** is an open position, access is provided to an input-output (I/O) interface **124** for connecting to or communicating with external devices as described below and to a battery receptacle **126** for placement and replacement of a battery (not shown). The battery receptacle **126** includes operative connections (not shown) for power transfer between the battery and the image capture device **100**. When the door **114** is in a closed position, the seal **120** engages a flange (not shown) or other interface to provide an environmental seal, and the battery interface **122** engages the battery to secure the battery in the battery receptacle **126**. The door **114** can also have a removed position (not shown) where the entire door **114** is separated from the image capture device **100**, that is, where both the hinge mechanism **116** and the latch mechanism **118** are decoupled from the body **102** to allow the door **114** to be removed from the image capture device **100**.
(15) The image capture device **100** may include a microphone **128** on a front surface and another microphone **130** on a side surface. The image capture device **100** may include other microphones on other surfaces (not shown). The microphones **128**, **130** may be configured to receive and record audio signals in conjunction with recording video or separate from recording of video. The image capture device **100** may include a speaker **132** on a bottom surface of the image capture device **100**. The image capture device **100** may include other speakers on other surfaces (not shown). The speaker **132** may be configured to play back recorded audio or emit sounds associated with notifications.
(16) A front surface of the image capture device **100** may include a drainage channel **134**. A bottom surface of the image capture device **100** may include an interconnect mechanism **136** for connecting the image capture device **100** to a handle grip or other securing device. In the example shown in FIG. **1**B, the interconnect mechanism **136** includes folding protrusions configured to move between a nested or collapsed position as shown and an extended or open position (not shown) that facilitates coupling of the protrusions to mating protrusions of other devices such as handle grips, mounts, clips, or like devices.
(17) The image capture device **100** may include an interactive display **138** that allows for interaction with the image capture device **100** while simultaneously displaying information on a surface of the image capture device **100**.
(18) The image capture device **100** of FIGS. **1**A-**1**B includes an exterior that encompasses and protects internal electronics. In the present example, the exterior includes six surfaces (i.e. a front face, a left face, a right face, a back face, a top face, and a bottom face) that form a rectangular cuboid. Furthermore, both the front and rear surfaces of the image capture device **100** are rectangular. In other embodiments, the exterior may have a different shape. The image capture device **100** may be made of a rigid material such as plastic, aluminum, steel, or fiberglass. The image capture device **100** may include features other than those described here. For example, the image capture device **100** may include additional buttons or different interface features, such as interchangeable lenses, cold shoes, and hot shoes that can add functional features to the image capture device **100**.
(19) The image capture device **100** may include various types of image sensors, such as charge-coupled device (CCD) sensors, active pixel sensors (APS), complementary metal-oxide-semiconductor (CMOS) sensors, N-type metal-oxide-semiconductor (NMOS) sensors, and/or any other image sensor or combination of image sensors.
(20) Although not illustrated, in various embodiments, the image capture device **100** may include other additional electrical components (e.g., an image processor, camera system-on-chip (SoC), etc.), which may be included on one or more circuit boards within the body **102** of the image capture device **100**.
(21) The image capture device **100** may interface with or communicate with an external device, such as an external user interface device (not shown), via a wired or wireless computing communication link (e.g., the I/O interface **124**). Any number of computing communication links

may be used. The computing communication link may be a direct computing communication link or an indirect computing communication link, such as a link including another device or a network, such as the internet, may be used.
(22) In some implementations, the computing communication link may be a Wi-Fi link, an infrared link, a Bluetooth (BT) link, a cellular link, a ZigBee link, a near field communications (NFC) link, such as an ISO/IEC 20643 protocol link, an Advanced Network Technology interoperability (ANT+) link, and/or any other wireless communications link or combination of links.
(23) In some implementations, the computing communication link may be an HDMI link, a USB link, a digital video interface link, a display port interface link, such as a Video Electronics Standards Association (VESA) digital display interface link, an Ethernet link, a Thunderbolt link, and/or other wired computing communication link.
(24) The image capture device **100** may transmit images, such as panoramic images, or portions thereof, to the external user interface device via the computing communication link, and the external user interface device may store, process, display, or a combination thereof the panoramic images.
(25) The external user interface device may be a computing device, such as a smartphone, a tablet computer, a phablet, a smart watch, a portable computer, personal computing device, and/or another device or combination of devices configured to receive user input, communicate information with the image capture device **100** via the computing communication link, or receive user input and communicate information with the image capture device **100** via the computing communication link.
(26) The external user interface device may display, or otherwise present, content, such as images or video, acquired by the image capture device **100**. For example, a display of the external user interface device may be a viewport into the three-dimensional space represented by the panoramic images or video captured or created by the image capture device **100**.
(27) The external user interface device may communicate information, such as metadata, to the image capture device **100**. For example, the external user interface device may send orientation information of the external user interface device with respect to a defined coordinate system to the image capture device **100**, such that the image capture device **100** may determine an orientation of the external user interface device relative to the image capture device **100**.
(28) Based on the determined orientation, the image capture device **100** may identify a portion of the panoramic images or video captured by the image capture device **100** for the image capture device **100** to send to the external user interface device for presentation as the viewport. In some implementations, based on the determined orientation, the image capture device **100** may determine the location of the external user interface device and/or the dimensions for viewing of a portion of the panoramic images or video.
(29) The external user interface device may implement or execute one or more applications to manage or control the image capture device **100**. For example, the external user interface device may include an application for controlling camera configuration, video acquisition, video display, or any other configurable or controllable aspect of the image capture device **100**.
(30) The user interface device, such as via an application, may generate and share, such as via a cloud-based or social media service, one or more images, or short video clips, such as in response to user input. In some implementations, the external user interface device, such as via an application, may remotely control the image capture device **100** such as in response to user input.
(31) The external user interface device, such as via an application, may display unprocessed or minimally processed images or video captured by the image capture device **100** contemporaneously with capturing the images or video by the image capture device **100**, such as for shot framing or live preview, and which may be performed in response to user input. In some implementations, the external user interface device, such as via an application, may mark one or more key moments contemporaneously with capturing the images or video by the image capture device **100**, such as

with a tag or highlight in response to a user input or user gesture.
(32) The external user interface device, such as via an application, may display or otherwise present marks or tags associated with images or video, such as in response to user input. For example, marks may be presented in a camera roll application for location review and/or playback of video highlights.
(33) The external user interface device, such as via an application, may wirelessly control camera software, hardware, or both. For example, the external user interface device may include a web-based graphical interface accessible by a user for selecting a live or previously recorded video stream from the image capture device **100** for display on the external user interface device.
(34) The external user interface device may receive information indicating a user setting, such as an image resolution setting (e.g., 3840 pixels by 2160 pixels), a frame rate setting (e.g., 60 frames per second (fps)), a location setting, and/or a context setting, which may indicate an activity, such as mountain biking, in response to user input, and may communicate the settings, or related information, to the image capture device **100**.
(35) The image capture device **100** may be used to implement some or all of the techniques described in this disclosure, such as the technique of sound removal shown and described in FIGS. **4**A-**6**.
(36) FIGS. **2**A-**2**B illustrate another example of an image capture device **200**. The image capture device **200** includes a body **202** and two camera lenses **204** and **206** disposed on opposing surfaces of the body **202**, for example, in a back-to-back configuration, Janus configuration, or offset Janus configuration. The body **202** of the image capture device **200** may be made of a rigid material such as plastic, aluminum, steel, or fiberglass.
(37) The image capture device **200** includes various indicators on the front of the surface of the body **202** (such as LEDs, displays, and the like), various input mechanisms (such as buttons, switches, and touch-screen mechanisms), and electronics (e.g., imaging electronics, power electronics, etc.) internal to the body **202** that are configured to support image capture via the two camera lenses **204** and **206** and/or perform other imaging functions.
(38) The image capture device **200** includes various indicators, for example, LEDs **208**, **210** to indicate a status of the image capture device **100**. The image capture device **200** may include a mode button **212** and a shutter button **214** configured to allow a user of the image capture device **200** to interact with the image capture device **200**, to turn the image capture device **200** on, and to otherwise configure the operating mode of the image capture device **200**. It should be appreciated, however, that, in alternate embodiments, the image capture device **200** may include additional buttons or inputs to support and/or control additional functionality.
(39) The image capture device **200** may include an interconnect mechanism **216** for connecting the image capture device **200** to a handle grip or other securing device. In the example shown in FIGS. **2**A and **2**B, the interconnect mechanism **216** includes folding protrusions configured to move between a nested or collapsed position (not shown) and an extended or open position as shown that facilitates coupling of the protrusions to mating protrusions of other devices such as handle grips, mounts, clips, or like devices.
(40) The image capture device **200** may include audio components **218**, **220**, **222** such as microphones configured to receive and record audio signals (e.g., voice or other audio commands) in conjunction with recording video. The audio component **218**, **220**, **222** can also be configured to play back audio signals or provide notifications or alerts, for example, using speakers. Placement of the audio components **218**, **220**, **222** may be on one or more of several surfaces of the image capture device **200**. In the example of FIGS. **2**A and **2**B, the image capture device **200** includes three audio components **218**, **220**, **222**, with the audio component **218** on a front surface, the audio component **220** on a side surface, and the audio component **222** on a back surface of the image capture device **200**. Other numbers and configurations for the audio components are also possible.
(41) The image capture device **200** may include an interactive display **224** that allows for

interaction with the image capture device **200** while simultaneously displaying information on a surface of the image capture device **200**. The interactive display **224** may include an I/O interface, receive touch inputs, display image information during video capture, and/or provide status information to a user. The status information provided by the interactive display **224** may include battery power level, memory card capacity, time elapsed for a recorded video, etc.

(42) The image capture device **200** may include a release mechanism **225** that receives a user input to in order to change a position of a door (not shown) of the image capture device **200**. The release mechanism **225** may be used to open the door (not shown) in order to access a battery, a battery receptacle, an I/O interface, a memory card interface, etc. (not shown) that are similar to components described in respect to the image capture device **100** of FIGS. **1**A and **1**B.

(43) In some embodiments, the image capture device **200** described herein includes features other than those described. For example, instead of the I/O interface and the interactive display **224**, the image capture device **200** may include additional interfaces or different interface features. For example, the image capture device **200** may include additional buttons or different interface features, such as interchangeable lenses, cold shoes, and hot shoes that can add functional features to the image capture device **200**.

(44) FIG. **3** is a block diagram of electronic components in an image capture device **300**. The image capture device **300** may be a single-lens image capture device, a multi-lens image capture device, or variations thereof, including an image capture device with multiple capabilities such as use of interchangeable integrated sensor lens assemblies. The description of the image capture device **300** is also applicable to the image capture devices **100**, **200** of FIGS. **1**A-**1**B and **2**A-**2**B.

(45) The image capture device **300** includes a body **302** which includes electronic components such as capture components **310**, a processing apparatus **320**, data interface components **330**, movement sensors **340**, power components **350**, and/or user interface components **360**.

(46) The capture components **310** include one or more image sensors **312** for capturing images and one or more microphones **314** for capturing audio.

(47) The image sensor(s) **312** is configured to detect light of a certain spectrum (e.g., the visible spectrum or the infrared spectrum) and convey information constituting an image as electrical signals (e.g., analog or digital signals). The image sensor(s) **312** detects light incident through a lens coupled or connected to the body **302**. The image sensor(s) **312** may be any suitable type of image sensor, such as a charge-coupled device (CCD) sensor, active pixel sensor (APS), complementary metal-oxide-semiconductor (CMOS) sensor, N-type metal-oxide-semiconductor (NMOS) sensor, and/or any other image sensor or combination of image sensors. Image signals from the image sensor(s) **312** may be passed to other electronic components of the image capture device **300** via a bus **380**, such as to the processing apparatus **320**. In some implementations, the image sensor(s) **312** includes a digital-to-analog converter. A multi-lens variation of the image capture device **300** can include multiple image sensors **312**.

(48) The microphone(s) **314** is configured to detect sound, which may be recorded in conjunction with capturing images to form a video. The microphone(s) **314** may also detect sound in order to receive audible commands to control the image capture device **300**.

(49) The processing apparatus **320** may be configured to perform image signal processing (e.g., filtering, tone mapping, stitching, and/or encoding) to generate output images based on image data from the image sensor(s) **312**. The processing apparatus **320** may include one or more processors having single or multiple processing cores. In some implementations, the processing apparatus **320** may include an application specific integrated circuit (ASIC). For example, the processing apparatus **320** may include a custom image signal processor. The processing apparatus **320** may exchange data (e.g., image data) with other components of the image capture device **300**, such as the image sensor(s) **312**, via the bus **380**.

(50) The processing apparatus **320** may include memory, such as a random-access memory (RAM) device, flash memory, or another suitable type of storage device, such as a non-transitory computer-

readable memory. The memory of the processing apparatus **320** may include executable instructions and data that can be accessed by one or more processors of the processing apparatus **320**. For example, the processing apparatus **320** may include one or more dynamic random-access memory (DRAM) modules, such as double data rate synchronous dynamic random-access memory (DDR SDRAM). In some implementations, the processing apparatus **320** may include a digital signal processor (DSP). More than one processing apparatus may also be present or associated with the image capture device **300**.

(51) The data interface components **330** enable communication between the image capture device **300** and other electronic devices, such as a remote control, a smartphone, a tablet computer, a laptop computer, a desktop computer, or a storage device. For example, the data interface components **330** may be used to receive commands to operate the image capture device **300**, transfer image data to other electronic devices, and/or transfer other signals or information to and from the image capture device **300**. The data interface components **330** may be configured for wired and/or wireless communication. For example, the data interface components **330** may include an I/O interface **332** that provides wired communication for the image capture device, which may be a USB interface (e.g., USB type-C), a high-definition multimedia interface (HDMI), or a FireWire interface. The data interface components **330** may include a wireless data interface **334** that provides wireless communication for the image capture device **300**, such as a Bluetooth interface, a ZigBee interface, and/or a Wi-Fi interface. The data interface components **330** may include a storage interface **336**, such as a memory card slot configured to receive and operatively couple to a storage device (e.g., a memory card) for data transfer with the image capture device **300** (e.g., for storing captured images and/or recorded audio and video).

(52) The movement sensors **340** may detect the position and movement of the image capture device **300**. The movement sensors **340** may include a position sensor **342**, an accelerometer **344**, or a gyroscope **346**. The position sensor **342**, such as a global positioning system (GPS) sensor, is used to determine a position of the image capture device **300**. The accelerometer **344**, such as a three-axis accelerometer, measures linear motion (e.g., linear acceleration) of the image capture device **300**. The gyroscope **346**, such as a three-axis gyroscope, measures rotational motion (e.g., rate of rotation) of the image capture device **300**. Other types of movement sensors **340** may also be present or associated with the image capture device **300**.

(53) The power components **350** may receive, store, and/or provide power for operating the image capture device **300**. The power components **350** may include a battery interface **352** and a battery **354**. The battery interface **352** operatively couples to the battery **354**, for example, with conductive contacts to transfer power from the battery **354** to the other electronic components of the image capture device **300**. The power components **350** may also include an external interface **356**, and the power components **350** may, via the external interface **356**, receive power from an external source, such as a wall plug or external battery, for operating the image capture device **300** and/or charging the battery **354** of the image capture device **300**. In some implementations, the external interface **356** may be the I/O interface **332**. In such an implementation, the I/O interface **332** may enable the power components **350** to receive power from an external source over a wired data interface component (e.g., a USB type-C cable).

(54) The user interface components **360** may allow the user to interact with the image capture device **300**, for example, providing outputs to the user and receiving inputs from the user. The user interface components **360** may include visual output components **362** to visually communicate information and/or present captured images to the user. The visual output components **362** may include one or more lights **364** and/or more displays **366**. The display(s) **366** may be configured as a touch screen that receives inputs from the user. The user interface components **360** may also include one or more speakers **368**. The speaker(s) **368** can function as an audio output component that audibly communicates information and/or presents recorded audio to the user. The user interface components **360** may also include one or more physical input interfaces **370** that are

physically manipulated by the user to provide input to the image capture device **300**. The physical input interfaces **370** may, for example, be configured as buttons, toggles, or switches. The user interface components **360** may also be considered to include the microphone(s) **314**, as indicated in dotted line, and the microphone(s) **314** may function to receive audio inputs from the user, such as voice commands.

(55) The image capture device **300** may be used to implement some or all of the techniques described in this disclosure, such as the sound removal technique described in FIGS. **4**A-**6**.

(56) FIGS. **4**A and **4**B illustrate block diagrams of processing a sound recording. The diagram of FIG. **4**A begins by initiating an audio capture **400**. During the audio capture **400**, a sound **402** is produced, for example, by an image capture device that is performing the audio capture **400** while also capturing video, such as the image capture devices **100**, **200**, **300** taught herein.

(57) The sound **402** indicates to a user that the device is on, the device is recording, or both. The sound **402** may be a beep, a whistle, a chime, a noise within a predetermined decibel range, a noise within a predetermined frequency range, have a predetermined frequency, or a combination thereof. The sound **402** may be created continuously or intermittently. The sound **402** may occur every 2 seconds or more, 5 seconds or more, 10 seconds or more, 15 seconds or more, or every 20 seconds or more. The sound **402** may occur every 5 minutes or less, 3 minutes or less, 1 minute or less, or 30 seconds or less. A user may select a frequency of the sound **402**. A user may select a tone, decibel, frequency, or a combination thereof of the sound **402**. A recording duration may be selected and the sound **402**, frequency of the sound **402**, or both may change as an end of the recording duration approaches. When the sound **402** is made, being prepared to be made, or has been made, the device (and/or apparatus) initiates a sound removal **404**.

(58) The sound removal **404** and the sound **402** production may be initiated at substantially the same time. The sound removal **404** may be initiated before the sound **402** is produced. The sound removal **404** may remove the sound **402** from an audio recording associated with a video recording captured by the device after the sound **402** is produced. The sound removal **404** may last a predetermined amount of time. The sound removal **404** may last a duration that is substantially a same duration as the sound **402**. The sound removal **404** may have a duration that is greater than a duration of the sound **402** production so that the sound **402** is not present in the audio recording. The sound removal **404** may use any process, processing, or device that is sufficient to remove the sound **402** from the audio recording or to prevent the sound **402** from becoming a part of the audio recording.

(59) The sound removal **404** may include applying a filter to the audio. The filter may be applied before the audio is captured, recorded, stored, or a combination thereof. The sound removal **404** may include applying a filter to remove any audio within a specific frequency range (e.g., including the sound **402**). The sound removal **404** may include recording sounds that are outside of a predetermined frequency without recording sounds within the predetermined frequency (e.g., the sound **402**). The sound removal **404** may produce tones configured to cancel the sound **402** at one or more microphones of the system. The sound removal **404** may include applying a notch filter, a frequency filter, cancellation by summation of antiphase signals, or both.

(60) For example, the sound removal **404** may apply an inverse filter to an audio signal that removes the sound **402** from the audio signal. The inverse filter may be configured to cancel the sound **402** during sound removal **404**. The inverse filter may be configured to apply a filter to a fundamental sound, a harmonic sound(s), or both so that the sound **402** is removed during sound removal **404**. The sound removal **404** may apply to one or more channels, two or more channels, or three or more channels. When more than one channel is present, the sound removal **404** may apply to all the channels or each channel may individually include the sound removal **404** (e.g., three separate filters). The processor may include or perform one or more sound removals **404**, two or more sound removals **404**, or three or more sound removals **404**. The sound removal **404** may only affect the sound **402** and may allow a remainder of the audio to remain unchanged. The sound

removal **404** may be applied to the microphones individually or all the microphones at one time. The sound removal **404** may only be performed electronically. The sound removal **404** may be performed by a processor. The processor discussed herein may apply the sound removal **404**. The processor may be configured to manipulate the sound **402** so that the sound is filtered.

(61) The sound removal **404** may include a mechanical filter. For example, the mechanical filter may be a microelectromechanical filter (MEMS filter). The mechanical filter may be applied during sound removal **404** to muffle the sound **402** at the microphone. The sound removal **404** may occur for a predetermined amount of time or during the duration of the sound **402** until sound completion **406**.

(62) The sound completion **406** may terminate the sound **402** after a predetermined amount of time. The sound completion **406** may end the sound **402** so that normal audio recording resumes without the sound **402** being produced. The sound **402** from start to sound completion **406** may last 3 seconds or less, 2 seconds or less, 1 second or less, 1 decisecond or less, 1 centisecond or less, or 1 millisecond or less. The sound **402** from start to sound completion **406** last 1 picoseconds or more, 5 picoseconds or more, 1 nanosecond or more, 5 nanoseconds or more, 1 microsecond or more, 5 microseconds or more, or 1 millisecond or more. Once the sound completion **406** occurs the sound removal is stopped **408**.

(63) The stopping **408** of the sound removal permits all audio (e.g., noises) to be captured, recorded, obtained, or a combination thereof by the image capture device. The stopping **408** turns the filters off so that the audio recorded is unmanipulated. The stopping **408** returns the audio recording to an untouched recording state (e.g., all raw audio is captured). The audio may be retouched (e.g., noise limited). The audio may be retouched during noise removal, normal recording, or both. The stopping **408** remains until another sound **402** is scheduled to begin and the process of removing the sound **402** is repeated.

(64) FIG. **4**B begins by initiating an audio capture **400** with a microphone. The audio capture **400** may be captured with one or more microphones. The audio capture **400** may be performed after selecting between multiple microphones and only recording from one of the microphones. The audio capture **400** may begin when record or capture is pressed, for example, by a user. The audio capture **400** may occur as long as a video is being recorded. The audio capture **400** may be unfiltered, untouched, or both until a sound **402** is programmed to occur.

(65) The sound **402** functions to indicate that a recording is occurring. The sound **402** is configured to indicate to a user that a video is being taken when the user is not able to see the image capture device such as the image capture devices **100**, **200**, **300**. The sound **402** may be within a predetermined frequency range, have a predetermined duration, occur at a predetermined time, or a combination thereof. The sound **402** may be beeps, tones, have a specific frequency, or a combination thereof. The sound **402** may be any sound taught herein. The sound **402** may have a predetermined fundamental sound, harmonic sound, or both. The sound **402** may be within an audible range for most humans. The sound **402** may be initiated concurrently with or made after sound removal **404** is initiated.

(66) The sound removal **404** may be turned on and then the sound **402** may occur so that the sound **402** is able to be removed, filtered out, or both (e.g., isolated) as the sound **402** is being made. The sound removal **404** may cancel the sound **402** such that the sound **402** is not recorded, stored in memory, or both (e.g., the audio or recording is free of the sound). The sound removal **404** may only affect the sound **402** and may not affect the other noises being recorded by image capture device. The sound removal **404** may only remove sounds **402** with a predetermined sound, frequency, duration, or a combination thereof. The sound removal **404** may begin before the sound **402** and continue until the sound completion **406**.

(67) The sound completion **406** may occur after a predetermined amount of time. The sound completion **406** may trigger the sound removal **404** to be stopped **408**. When the sound removal **404** is stopped **408**, normal recording (e.g., of both audio and video) resumes. When the sound

removal **404** is stopped **408**, the audio capture **400** may continue to record audio without a filter, without sound removal, or both.
(68) FIGS. **5**A-**5**C illustrate graphs of an audio recording and the audio recording being filtered. FIG. **5**A illustrates an unfiltered microphone diagram sound curve **500**. The sound curve **500** begins recording before a sound is initiated and then the sound occurs. As shown, the sound includes a fundamental sound **502** and harmonic sounds **504**. The fundamental sound **502** may be an initial sound that is received by a microphone (or two or more microphones). The fundamental sound **502** may be a normal tone that is produced by a speaker and captured by a microphone. The fundamental sound **502** may be a first sound recorded by a microphone. The fundamental sound **502** may be a primary sound or a loudest sound amount multiple sounds. The fundamental sound **502** may be a first sound recorded and then the harmonic sounds **504** may be recorded second or captured second.
(69) The harmonic sounds **504** may be a non-ideal sound, sound distortion, reverberated sound, delayed sound, echo, or a combination thereof. The harmonic sounds **504** may have a different frequency than the fundamental sound **502**. If more than one harmonic sound **504** is present, each harmonic sound **504** may increase in frequency approximately double (e.g., by adding an amount of frequency of the fundamental sound **502** together for each successive frequency of the harmonic sound **504**). For example, as shown in FIG. **5**A, the fundamental sound **502** has a frequency of about 3000 Hz, a first harmonic sound **504** has a frequency of about 6000 Hz, and a second harmonic sound **504** has a frequency of about 9000 Hz. As shown, the fundamental sound **502** and the harmonic sound **504** are ascertained and then a filter may be applied. The fundamental sound **502** and the harmonic sound **504** may be ascertained by selecting a predetermined sound. The device may be calibrated to ascertain the fundamental sound **502**, the harmonic sound **504**, or both. The device may be pre-programmed with the fundamental sound **502**, the harmonic sound **504**, or both. The device upon an initial turning on may play the sound and capture the sound to ascertain the fundamental sound **502**, the harmonic sound **504**, or both.
(70) The sounds may be calibrated so that the fundamental sound **502** and the harmonic sounds **504** are in a predetermined range, at specific frequency, at specific decibel level, have a predetermined strength, have a predetermined characteristic, or a combination thereof.
(71) FIG. **5**B illustrates a filter curve **510**. The filter curve **510** applies an inverse signal to the fundamental sound **502** and the harmonic sounds **504** so that the fundamental sound **502** and the harmonic sounds **504** are cancelled and/or removed. As shown, a fundamental filter **512** and harmonic filters **514** are applied that are commensurate in inverse size to the fundamental sound **502** and the harmonic sounds **504**, respectively. For example, FIG. **5**A shows that the fundamental sound **502** has a frequency of 3000 Hz and a volume of about 45 decibels above the nominal level (e.g., frequencies of surrounding sounds) and the fundamental filter **512** is applied around 3000 Hz and about-45 decibels such that the fundamental filter **512** may cancel substantially all (e.g., 90 percent or more or 95 percent or more) of the fundamental sound **502**.
(72) The fundamental filter **512**, the harmonic filters **514**, or both may essentially proportionally and/or inversely match the respective fundamental sound **502**, the harmonic sounds **504**, or both such that the fundamental sound **502**, the harmonic sounds **504**, or both are essentially completely removed (e.g., 80 percent or more, 90 percent or more, or even 95 percent or more of the sound is removed). The fundamental filter **512**, the harmonic filters **514**, or both are applied by a process, performed within a processor, or both.
(73) FIG. **5**C illustrates the filtered sound curve **520** when the filter curve **510** of FIG. **5**B is applied to the sound curve **500** of FIG. **5**A. As illustrated, the fundamental sound **502** and the harmonic sounds **504** are absent from the filtered sound curve **520**.
(74) FIG. **6** illustrates an example of a sound and microphone diagram **600**. The sound and microphone diagram **600** illustrates a first sound **602** from a first location (e.g., a speaker of the image capture device, labeled “S”) and a second sound **604** from a second location (e.g., people or

a surrounding environment, labeled “E”). The first sound **602** hits a first microphone **606** (labeled “A”) first and then a second microphone **608** (labeled “C”) second. The first sound **602** is located a distance X **610** from the first microphone **606**. The first sound **602** is located a distance Y **612** from the second microphone **608**. The distance X **610** is smaller than that distance Y **612**. The distance X **610** and the distance Y **612** may be determined based upon a location of the microphones **606** and **608** on the image capture device **100**, **200**, **300**. For example, if the microphones **606** and **608** are located on the front and back (e.g., spaced by a thickness of the image capture device) the difference between the distance X **610** and the distance Y **612** (e.g., a distance apart) may be less than if the microphones **606** and **608** are located on a left side and a right side (e.g., spaced by a length of the image capture device) such as when the image capture device is similar to the image capture device **100**, **200**, or **300**.
(75) FIG. **6** illustrated leveraging multiple microphones (e.g., the first microphone **606** and the second microphone **608**) to remove the sound by subtracting one microphone from another given the known microphone and speaker placements. Filtration of an audio signal received by the first microphone **606** will occur and the audio signal received by the second microphone **608** will be filtered so that the sound is removed and the ambient sound is preserved (i.e., filtration will occur such as the filtration occurring removes the sound and the ambient sound is preserved). The delay may be calculated based upon a distance between the distance X **610** and the distance Y **612** and an amount of time it takes the first sound **602** to travel that distance.
(76) The first sound **602** may be a sound created by a speaker in the image capture device. The first sound **602** is monitored and filtered while the second sound **604** is being recorded and captured by the first microphone **606** and the second microphone **608**. Calibration may be applied to the image capture device so that the filter may be applied accurately to the first sound **602** based upon the difference between the first distance X **610** and the second distance Y **612**. The microphone **606**, **608** with the cleanest, loudest, or otherwise determined best audio may be selected and then the filter only applied to the selected microphone, e.g., either the first microphone **606** or the second microphone **608**.
(77) A step of calibrating may be applied universally to all image capture devices with a specific orientation of the first microphone **606** to the second microphone **608**. Calibration may have to be applied to every image capture device individually. Calibration may calibrate when the filter is applied, how long the filter is applied, an intensity or frequency of the audio to be recorded, a fundamental sound, a harmonic sound, or a combination thereof. The step of calibration may be performed by a processor taught herein. The processor may perform the processes taught herein.
(78) While the disclosure has been described in connection with certain embodiments, it is to be understood that the disclosure is not to be limited to the disclosed embodiments but, on the contrary, is intended to cover various modifications and equivalent arrangements included within the scope of the appended claims, which scope is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures as is permitted under the law.

Claims

1. An apparatus comprising: a speaker configured to produce a sound that audibly communicates information to a user; a microphone configured to receive the sound; and a processor configured to: cause the speaker to produce the sound to audibly communicate the information to the user; capture a video that includes audio; initiate a sound removal process to remove the sound from the audio; and after the sound is produced, stop the sound removal process, wherein the sound is a beep or a chime and the sound removal process lasts about 5 seconds or less.
2. The apparatus of claim 1, wherein the processor is further configured to produce a recording that includes the audio that is free of the sound.
3. The apparatus of claim 1, wherein the sound removal process to remove the sound includes a

notch filter.
4. The apparatus of claim 3, wherein the notch filter removes a fundamental sound and harmonics sounds.
5. The apparatus of claim 1, wherein the sound removal process to remove the sound includes a frequency filter.
6. The apparatus of claim 1, wherein the sound removal process removes the sound by filtering three channels with three separate filters.
7. The apparatus of claim 1, wherein the processor is further configured to calibrate the sound removal process for the apparatus to isolate characteristics of the sound, a strength of harmonics of the sound, or both, and wherein the processor is further configured to filter only sounds within a predetermined specific range.
8. The apparatus of claim 7, wherein the information audibly communicated relates to information displayed on an interactive display of the apparatus.
9. The apparatus of claim 1, wherein the microphone is two or more microphones that are located a distance apart and the sound is removed from the two or more microphones based upon the distance between the two or more microphones so that the sound is removed.
10. A method comprising: initiating a sound removal process; after initiation of the sound removal process, producing a sound that audibly communicates information to a user; upon completion of producing the sound, stopping the sound removal process; and producing a recording including audio that is free of the sound, wherein the sound is a beep or a chime and the sound removal process lasts about 5 seconds or less.
11. The method of claim 10, further comprising: applying a notch filter to the sound to remove a fundamental sound and a harmonic sound.
12. The method of claim 10, further comprising: displaying the information to the user via a touch screen of an image capture device while audibly communicating the information.
13. The method of claim 12, wherein the information comprises camera mode, battery life, or memory card capacity.
14. The method of claim 10, further comprising: applying a frequency filter, and filtering three audio channels with three separate filters.
15. The method of claim 14, further comprising: calibrating the sound removal process for an apparatus to isolate characteristics of the sound, a strength of harmonics of the sound, or both.
16. The method of claim 15, wherein the sound is within a predetermined specific frequency range and the method further comprises filtering out only sounds within the predetermined specific frequency range.
17. The method of claim 10, wherein removing the sound by subtracting a first microphone from a second microphone is based on a position of the first microphone with respect to the second microphone.
18. An image capture device comprising: a speaker configured to produce a sound that audibly communicates information to a user; a microphone configured to receive the sound and additional sounds; an image sensor configured to capture a video while the microphone is receiving the sound and the additional sounds; and a processor configured to: initiate a sound removal process to remove the sound from a recording that includes the sound and the additional sounds; produce the sound to audibly communicate the information; and stop the sound removal process after a predetermined amount of time, wherein the sound is a beep or a chime and the sound removal process lasts about 5 seconds or less.
19. The image capture device of claim 18, wherein the information audibly communicated relates to information displayed on an interactive display of the image capture device.
20. The image capture device of claim 18, wherein the processor is calibrated to ascertain a fundamental sound of the sound so that the sound is removed from the recording.